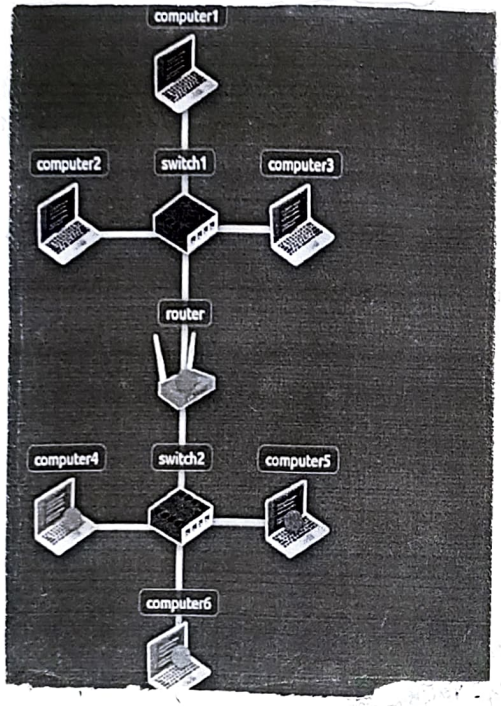
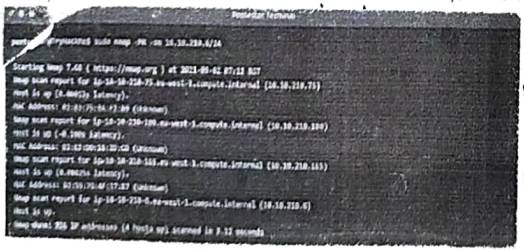




18/09/2025 Ex:08 NMAP to Discover Live Hosts Using Nmap Scans (ARP, ICMP, TCP/UDP) on the TryHackMe Platform

(a)

AIM:

To discover Live Hosts using Nmap Scans (ARP, ICMP, TCP/UDP) on the TryHackMe Platform Room Link: https://tryhackme.com/room/nmap01.

1. Introduction to Nmap:

Nmap is a free and open-source network scanning tool used to discover active systems and identify running services on a network.

Before performing port scanning, it is useful to determine which systems are actually online. Scanning offline system wastes time and can create unnecessary network traffic.

Nmap provides different methods for live host discovery, including:

* ARP scan
* ICMP scan
* TCP SYN ping.
* TCP ACK ping.
* UDP ping.

The exp is based on the Nmap live Host Discovery room of TryHackMe.

2. Subnetworks:

A network segment is a group of computers connected through a common medium such as an Ethernet switch / WiFi access point.

A subnetwork (subnet) is a logical portion of an IP network having it's own IP address range.

Ex:

Subnet	Subnet Mask	Approx. Hosts.
/16	255.255.0.0	~ 65,000
/24	255.255.255.0	~ 250.

ARP-based discovery works only within the same subnet, because ARP operates at the link layer and its requests cannot cross routers.

3. Enumerating Target:

Before scanning, Nmap needs to know which systems / IP addresses should be checked.

Targets can be specified as:

⇒ Single IP address: 10.10.10.5.
⇒ Range of IP addresses: 10.10.10.5-10
⇒ Subnet: 10.10.10.0/24.
⇒ List of targets from a file: nmap -iL lists-of-hosts.txt.
⇒ Display targets without scanning: nmap -sL TARGETS.
⇒ Disable DNS resolution: nmap -n TARGETS.

Thus, target enumeration helps define the scope of the scan.

4. Live Host Discovery:

The experiment covers 4 protocols:

ARP - Link Layer.
ICMP - Network layer.
TCP - Transport layer.
UDP - Transport layer.

5. ARP Host Discovery:

* ARP requests are used to find live hosts on the same subnet.
* A live host responds with an ARP reply.
* ARP cannot cross routers.
* nmap -PR -sn TARGETS performs an ARP-based host discovery without port scanning.
* In the TryHackMe activity, 3 devices were discovered using ARP requests.

6. ICMP Host Discovery:

* ICMP Echo uses Type 8 (Echo Request) and Type 0 (Echo Reply)
* Nmap option: -PE
* ICMP Timestamp scan: -PP.
* ICMP Address Mask Scan: -PM.
* ICMP may not always work because firewalls can block ICMP requests.

7. TCP SYN Ping:

* Sends a TCP packet with SYN flag set.
* An open port may respond with SYN/ACK
* A closed port may respond with RST
* Nmap option: -PS.

* Example for Telnet port 23: -PS23
* TCP SYN ping does not require a privileged account according to the manual's TryHackMe section.

8. TCP ACK ping:
* sends a TCP packet with the ACK flag set.
* Normally uses port 80 by default.
* Nmap option: -PA
* A target may respond with a RST, indicating that the host is active.
* TCP ACK ping requires a privileged account.

9. UDP ping:
* UDP packets can be used to determine whether a host is active.
* A packet sent to a closed UDP port may produce an ICMP "Port Unreachable" response.
* This response indicates that the target host is active and reachable.

10. Reverse DNS lookup:
* Nmap normally performs reverse-DNS lookups for online hosts.
* -n → disables DNS resolution.
* -R → performs reverse-DNS lookup even for offline hosts.
* A specific DNS server can also be specified using --dns-servers DNS SERVER.

	PURPOSE	option.
1.	Ping/host discover only	-sn
2.	ARP Scan	-PR
3.	ICMP Echo	-PE
4.	ICMP Timestamp	-PP
5.	ICMP Address Mask	-PM
6.	TCP SYN Ping	-PS
7.	TCP ACK ping	-PA
8.	Disable DNS lookup	-n
9.	Force reverse DNS	-R
10.	List targets without scanning	-sL
11.	Read targets from file	-iL

RESULT: The Nmap live Host Discovery experiment was successfully studied on the TryHackMe platform. Live hosts were identified using ARP, ICMP, TCP, ACK and UDP techniques and the corresponding Nmap options were understood.